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CSA1717 – ARTIFICIAL INTELLIGENCE

CO1 – LIST LAB PROGRAMS PYTHON

EXP 2: 8-Queen Problem

PROGRAM:

practical lab 2.py - C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 2.py (3.14.5)

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```
N = 8
board = [-1] * N

def safe(row, col):
    for r in range(row):
        if board[r] == col:
            return False
        if abs(board[r] - col) == abs(r - row):
            return False
    return True

def solve(row):
    if row == N:
        return True

    for col in range(N):
        if safe(row, col):
            board[row] = col

            if solve(row + 1):
                return True

            board[row] = -1

    return False

if solve(0):
    print("One solution to the 8-Queen problem:")
    for row in range(N):
        for col in range(N):
            if board[row] == col:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()
else:
    print("No solution")
```

OUTPUT:

IDLE Shell 3.14.5

File Edit Shell Debug Options Window Help

```
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 2.py ====
One solution to the 8-Queen problem:
Q . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
      Q
```